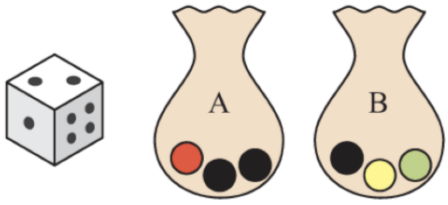
EXERCISE 1

[4 marks]

Michelle rolls a die. If the result is even, she picks a ball from bag A. Otherwise, she picks a ball from bag B.
How many ways lead to:

a a yellow ball being picked

b a black ball being picked?



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EXERCISE 2

[4 marks]

Express in factorial notation :

1. $\frac{1}{8 \times 7 \times 6 \times 5}$;
2. $\frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1}$.

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EXERCISE 3

[2 marks] Simplify :

1. $\frac{(n-1)!}{(n-2)!}$;
2. $\frac{(n+2)!}{n!}$.

EXERCISE 4

[2 marks]

Without multiplying all the terms, show that : $10! = 6!7!$

EXERCISE 5

[7 marks]

Sprinters A, B, C, and D are members of a 4×400 m relay team. In how many ways can the sprinters be ordered if :

1. there are no restrictions
2. B must sprint last
3. D must not sprint last
4. C must sprint either first or second
5. A must be followed directly by B

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EXERCISE 6

[2 marks]

In how many different ways can 4 identical red balls, 3 identical green balls and a yellow ball be arranged in a row ?

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